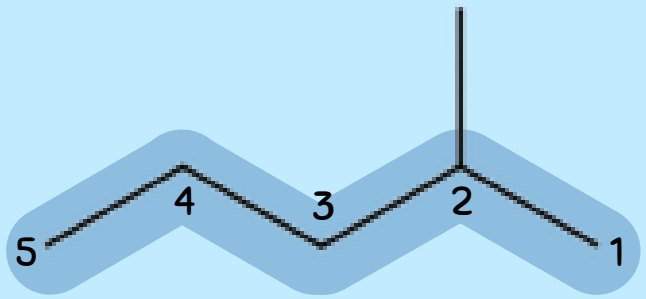
PART A · STRUCTURE → NAME
A01 · Foundation · C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 5 C, so the parent is pentane. The remaining carbon is a methyl branch.

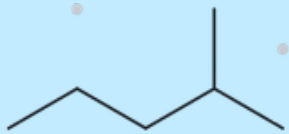
Numbering: Numbering from the left puts the methyl on C2. Numbering from the right puts it on C4. Since 2 is lower than 4, number from the left.

Substituents: One methyl on C2, so the name is 2-methylpentane.

3. My Name

2-methylpentane

4. The Alkane Check



Returned Name :

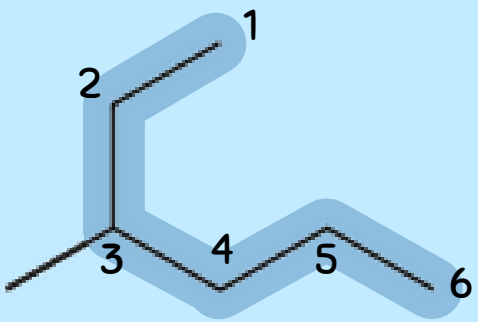
2-methylpentane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE → NAME
A02 · Foundation · C_7H_{16}

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 6 C, so the parent is hexane. The remaining carbon is a methyl branch.

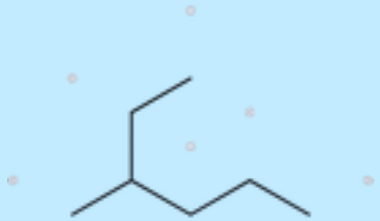
Numbering: Numbering from either end places the methyl on C3 or C4. Choose the direction that gives the lower number, C3.

Substituents: One methyl on C3, so the name is 3-methylhexane.

3. My Name

3-methylhexane

4. The Alkane Check



Returned Name :

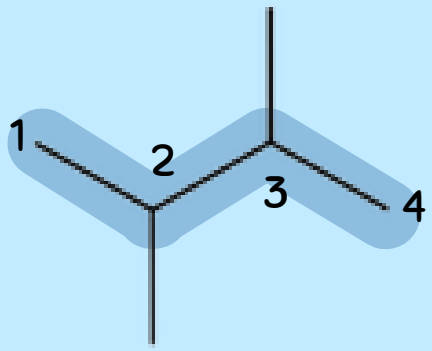
3-methylhexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➜ NAME
A03 · Foundation · C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 4 C, so the parent is butane. The remaining two carbons are methyl branches.

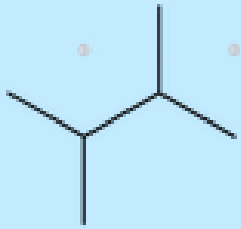
Numbering: Numbering from either end gives the same positions: C2 and C3. Therefore, either direction is acceptable.

Substituents: Two methyl groups, one on C2 and one on C3, so the name is 2,3-dimethylbutane.

3. My Name

2,3-dimethylbutane

4. The Alkane Check



Returned Name :

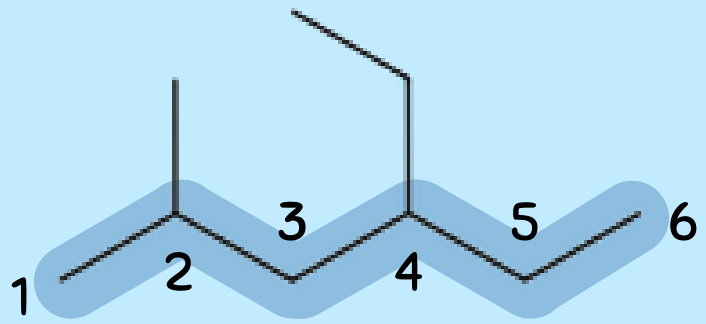
2,3-dimethylbutane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➔ NAME
A04 · Developing · C_9H_{20}

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 6 C, so the parent is hexane. The remaining branches are one ethyl group and one methyl group.

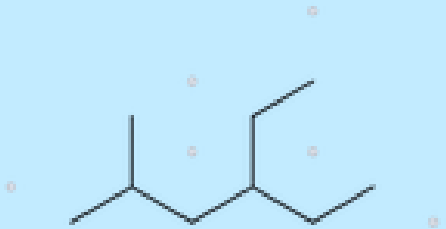
Numbering: Numbering from one end gives substituents on C2 and C4. Numbering from the opposite end gives C3 and C5. Since 2,4 is the lower set of locants, number from the end nearest the methyl branch.

Substituents: One methyl on C2 and one ethyl on C4. Arrange the substituents alphabetically, so the name is 4-ethyl-2-methylhexane.

3. My Name

4-ethyl-2-methylhexane

4. The Alkane Check



Returned Name :

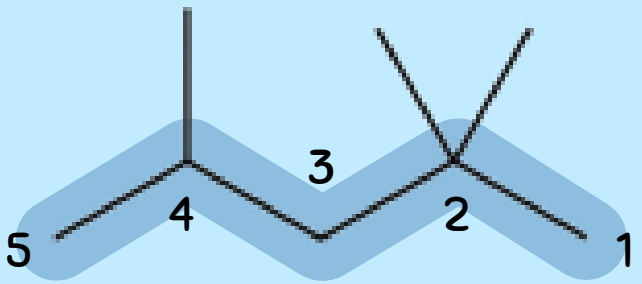
4-ethyl-2-methylhexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➜ NAME
A05 · Developing · C_8H_{18}

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 5 C, so the parent is pentane. The remaining three carbons are methyl branches.

Numbering: Numbering from one end gives methyl groups on C2, C2, and C4. Numbering from the opposite end gives C2, C4, and C4. The first set, 2,2,4, is lower than 2,4,4, so number from the first end.

Substituents: Two methyl groups on C2 and one methyl group on C4, so the name is 2,2,4–trimethylpentane.

3. My Name

2,2,4–trimethylpentane

4. The Alkane Check



Returned Name :

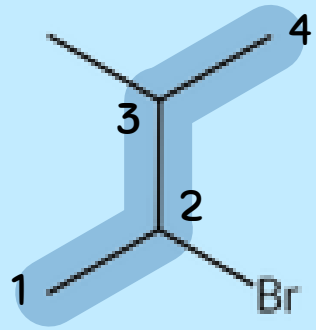
2,2,4-trimethylpentane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➜ NAME
A06 · Developing · $C_5H_{11}Br$

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 4 C, so the parent is butane. The remaining substituents are one bromo group and one methyl group.

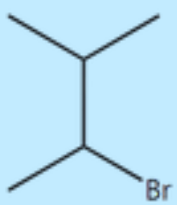
Numbering: Numbering from either end gives the locant set 2,3. Since there is a tie, give the lower number to the substituent that comes first alphabetically. Bromo comes before methyl, so bromine gets C2.

Substituents: One bromo on C2 and one methyl on C3, so the name is 2-bromo-3-methylbutane.

3. My Name

2-bromo-3-methylbutane

4. The Alkane Check



Returned Name :

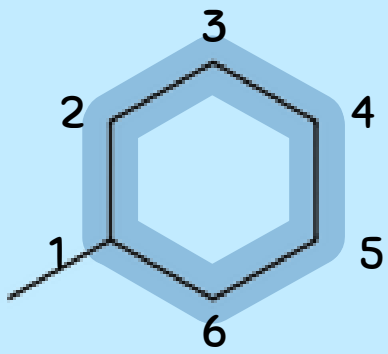
2-bromo-3-methylbutane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➔ NAME
A07 · Developing · C_7H_{14}

1. Numbered Parent



2. Naming Trail

Parent: The structure contains a 6–carbon ring, so the parent is cyclohexane. The ring is the parent because the activity specifies that the ring takes priority when a ring is present.

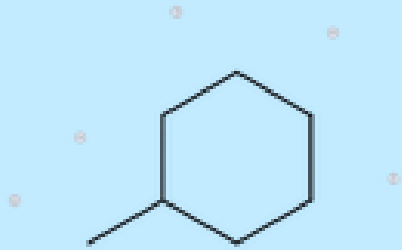
Numbering: The carbon attached to the methyl group is assigned C1. Since there is only one substituent, no other numbering direction needs to be considered.

Substituents: One methyl on C1. The name is methylcyclohexane.

3. My Name

methylcyclohexane

4. The Alkane Check



Returned Name :

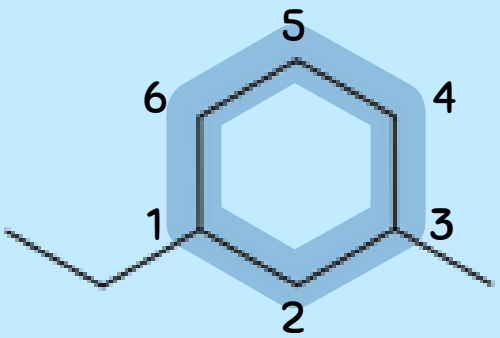
methylcyclohexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➜ NAME
A08 · Challenge · C_9H_{18}

1. Numbered Parent



2. Naming Trail

Parent: The structure contains a 6-carbon ring, so the parent is cyclohexane. The ethyl and methyl groups are attached to the ring.

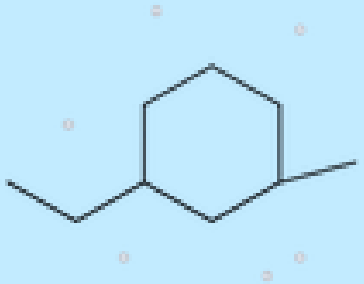
Numbering: Start numbering at the carbon attached to the ethyl group as C1. Number toward the methyl group to give it C3. This gives the lowest possible set of locants, 1,3.

Substituents: One ethyl on C1 and one methyl on C3. Arrange them alphabetically, so the name is 1-ethyl-3-methylcyclohexane.

3. My Name

1-ethyl-3-methylcyclohexane

4. The Alkane Check



Returned Name :

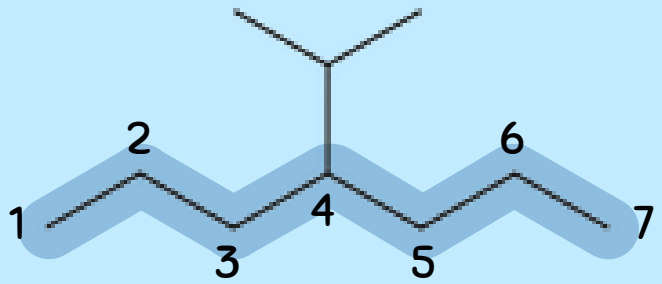
1-ethyl-3-methylcyclohexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➜ NAME
A09 · Challenge · $C_{10}H_{22}$

1. Numbered Parent



2. Naming Trail

Parent: The longest continuous chain is 7 C, so the parent is heptane. The remaining branch is a three-carbon branched substituent called 1-methylethyl.

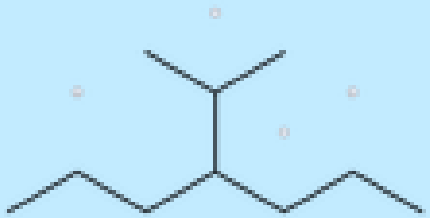
Numbering: Numbering from either end places the substituent on C4. Both directions give the same locant, so either direction is acceptable.

Substituents: One 1-methylethyl group on C4, so the name is 4-(1-methylethyl)heptane.

3. My Name

4-(1-methylethyl)heptane

4. The Alkane Check



Returned Name :

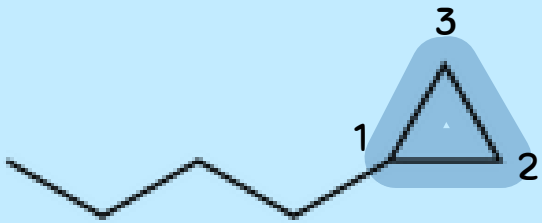
2-methyl-3-propylhexane

MIS-MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A · STRUCTURE ➜ NAME
A10 · Challenge · C_8H_{16}

1. Numbered Parent



2. Naming Trail

Parent: The structure contains a 3-carbon ring and a 5-carbon chain. Following the activity's rule that the ring is the parent when a ring is present, the parent is cyclopropane.

Numbering: The ring carbon attached to the pentyl group is assigned C1. Since there is only one substituent, no other numbering direction needs to be considered.

Substituents: One pentyl group on C1. The name is pentylcyclopropane.

3. My Name

butylcyclopropane

4. The Alkane Check



Returned Name :

butylcyclopropane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART B · NAME ➜ STRUCTURE
B01 · Foundation · C_6H_{14}

1. Name Decoded

Parent: pentane, a 5–carbon chain.

Substituents: methyl on C2.

Expected formula: 5+1 = 6 C. For an open–chain alkane, C_nH_{2n+2} gives 2(6) + 2 = 14 H, so C_6H_{14}.

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_6H_{14}

- 3 CH_3 (C1, C5 and the methyl branch) = 9 H
- 2 CH_2 (C3, C4) = 4 H
- 1 CH (C2) = 1 H
- Total = 14 H

Open chain: $C_nH_{2n+2} = C_6H_{14}$ ✓

4. Back Check

Returned by The Alkane:

2-methylpentane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back–check recorded

PART B · NAME ➜ STRUCTURE
B02 · Foundation · C_7H_{16}

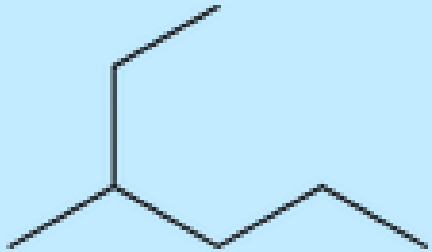
1. Name Decoded

Parent: hexane, a 6–carbon chain.

Substituents: methyl on C3.

Expected formula: 6 + 1 = 7 C. For an open–chain alkane, C_nH_{2n+2} gives 2(7) + 2 = 16 H, so C_7H_{16}.

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_6H_{14}

- 3 CH_3 (C1, C6 and the methyl branch) = 9 H
- 3 CH_2 (C2, C4, C5) = 6 H
- 1 CH (C3) = 1 H
- Total = 16 H

Open chain: $C_nH_{2n+2} = C_6H_{14}$ ✓

4. Back Check

Returned by The Alkane:

3-methylhexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back–check recorded

PART B · NAME ➜ STRUCTURE
B03 · Foundation · C_6H_{14}

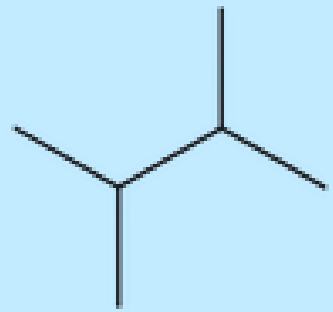
1. Name Decoded

Parent: butane, a 4–carbon chain.

Substituents: methyl on C2 and methyl on C3 (two methyls, so "di").

Expected formula: 4 + 2 = 6 C. For an open–chain alkane, C_nH_{2n+2} gives 2(6) + 2 = 14 H, so C_6H_{14}.

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_6H_{14}

- 4 CH_3 (C1, C4 and the two methyl branches) = 12 H
- 2 CH (C2, C3) = 2 H
- Total = 14 H

Open chain: $C_nH_{2n+2} = C_6H_{14}$ ✓

4. Back Check

Returned by The Alkane:

2,3-dimethylbutane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back–check recorded

PART B · NAME ➜ STRUCTURE
B04 · Developing · C_9H_{20}

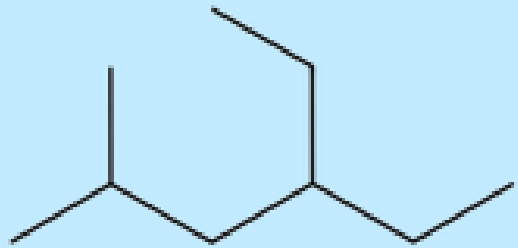
1. Name Decoded

Parent: hexane, a 6-carbon chain.

Substituents: methyl on C2; ethyl on C4 (listed alphabetically, ethyl before methyl).

Expected formula: 6 + 1 + 2 = 9 C. For an open-chain alkane, C_nH_{2n+2} gives 2(9) + 2 = 20 H, so C_9H_{20}.

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_9H_{20}

- 4 CH_3 (C1, C6, the methyl branch, the ethyl's end carbon) = 12 H
- 3 CH_2 (C3, C5, the ethyl's first carbon) = 6 H
- 2 CH (C2, C4) = 2 H
- Total = 20 H

Open chain: $C_nH_{2n+2} = C_9H_{20}$ ✓

4. Back Check

Returned by The Alkane:

4-ethyl-2-methylhexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B · NAME ➜ STRUCTURE
B05 · Developing · C_8H_{18}

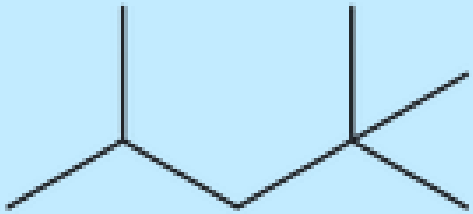
1. Name Decoded

Parent: pentane, a 5-carbon chain.

Substituents: two methyls on C2 and one methyl on C4 (three methyls, so "tri").

Expected formula: 5 + 3 = 8 C. For an open-chain alkane, C_nH_{2n+2} gives 2(8) + 2 = 18 H, so C_8H_{18}.

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_8H_{18}

- 5 CH_3 (C1, C5 and the three methyl branches) = 15 H
- 1 CH_2 (C3) = 2 H
- 1 CH (C4) = 1 H
- 1 C with no H (C2, quaternary) = 0 H
- Total = 18 H
- Open chain: $C_nH_{2n+2} = C_8H_{18}$ ✓

Carbon count: 5 + 1 + 1 + 1 = 8 C ✓

4. Back Check

Returned by The Alkane:

2,2,4-trimethylpentane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B · NAME → STRUCTURE
B06 · Developing · $C_5H_{11}Br$

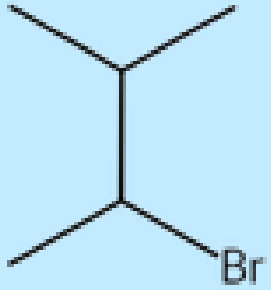
1. Name Decoded

Parent: butane → a 4-carbon chain.

Substituents: bromo on C2; methyl on C3

Expected formula: 4 + 1 = 5 C; one chain → C_5H_{12}; Br replaces one H → $C_5H_{11}Br$

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: $C_5H_{11}Br$

- 3 CH_3 (C1, C4, the methyl branch) = 9 H
- 2 CH (C2 and C3) = 2 H
- Total = 11 H, plus 1 Br

Open chain: C_nH_{2n+2} = C_5H_{12}; minus one H for Br = $C_5H_{11}Br$ ✓

4. Back Check

Returned by The Alkane:

2-bromo-3-methylbutane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B · NAME ➜ STRUCTURE
B07 · Developing · C_7H_{14}

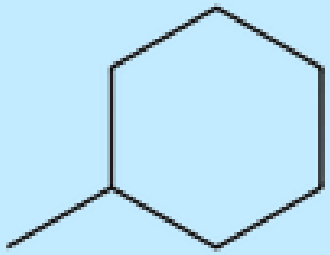
1. Name Decoded

Parent: cyclohexane → a 6-carbon ring.

Substituents: methyl on C1 (the locant is not written, since there is only one substituent).

Expected formula: 6 + 1 = 7 C; one ring → C_6H_{12}; the methyl replaces one H → C_7H_{14}

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_7H_{14}

- 1 CH (C1) = 1 H
- 5 CH_2 (C2 to C6) = 10 H
- 1 CH_3 (methyl) = 3 H
- Total = 14

One ring: $C_nH_{2n} = C_6H_{12}$; the methyl replaces one H and adds CH_3, so $C_6H_{11} + CH_3 = C_7H_{14}$ ✓

4. Back Check

Returned by The Alkane:

methylcyclohexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B · NAME ➜ STRUCTURE
B08 · Challenge · C_9H_{18}

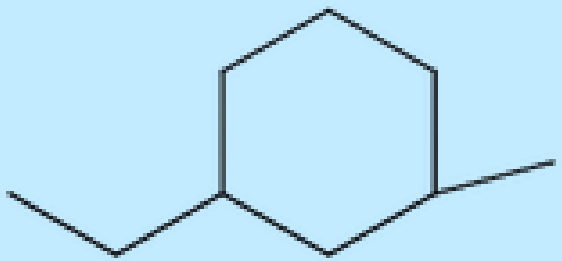
1. Name Decoded

Parent: cyclohexane → a 6-carbon ring.

Substituents: ethyl on C1; methyl on C3 (listed alphabetically).

Expected formula: 6 + 2 + 1 = 9 C; one ring → C_6H_{12}; ethyl and methyl each replace one H → C_9H_{18}

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_9H_{18}

- 2 ring CH (C1, C3) = 2 H
- 4 ring CH_2 (C2, C4, C5, C6) = 8 H
- 1 ethyl CH_2 = 2 H
- 2 CH_3 (ethyl end, methyl) = 6 H
- Total = 18 H

Open chain: One ring: $C_nH_{2n} = C_6H_{12}$; minus two H for the substituents = C_6H_{10}, plus C_2H_5 and CH_3 = C_9H_{18} ✓

4. Back Check

Returned by The Alkane:

1-ethyl-3-methylcyclohexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B · NAME ➜ STRUCTURE
B09 · Challenge · $C_{10}H_{22}$

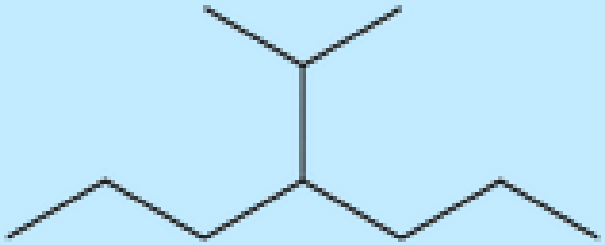
1. Name Decoded

Parent: heptane → a 7-carbon chain.

Substituents: 1-methylethyl (isopropyl) on C4.

Expected formula: 7 + 3 = 10 C; open chain → C_7H_{16}; the branch replaces one H → $C_{10}H_{22}$

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: $C_{10}H_{22}$

- 4 CH_3 (C1, C7, two isopropyl methyls) = 12 H
- 4 CH_2 (C2, C3, C5, C6) = 8 H
- 2 CH (C4, isopropyl CH) = 2 H
- Total = 22 H

Open chain: $C_nH_{2n+2} = C_{10}H_{22}$ ✓

4. Back Check

Returned by The Alkane:

2-methyl-3-propylhexane

Not the same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B · NAME ➜ STRUCTURE
B10 · Challenge · C_7H_{14}

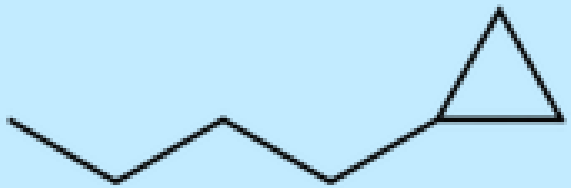
1. Name Decoded

Parent: cyclopropane → a 3-carbon ring.

Substituents: butyl on C1 (the locant is not written, since there is only one substituent).

Expected formula: 3 + 4 = 7 C; one ring → C_3H_6; butyl replaces one H → C_7H_{14}

2. Bond – Line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_7H_{14}

- 1 ring CH (C1) = 1 H
- 2 ring CH_2 (C2, C3) = 4 H
- 3 chain CH_2 = 6 H
- 1 CH_3 = 3 H
- Total = 14 H

One ring: $C_nH_{2n} = C_3H_6$; butyl replaces one H, so C_3H_5 + $C_4H_9 = C_7H_{14}$ ✓

4. Back Check

Returned by The Alkane:

butylcyclopropane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded